

1. LSM-Tree Compaction Architecture

HyperStore DB utilizes a Log-Structured Merge (LSM) storage engine optimized for high-throughput write workloads. MemTables are flushed to Level-0 SSTables when active buffer size reaches 64 MB. Level-0 allows up to 4 overlapping files before triggering Level-0 to Level-1 compaction.

2. Compaction Size Multipliers

Each level from Level-1 upward uses a 10x size amplification multiplier. Level-1 capacity is capped at 100 MB, Level-2 at 1,000 MB (1 GB), and Level-3 at 10 GB.

3. Write-Ahead Log (WAL) Durability

All write transactions append to a synchronous Write-Ahead Log. Under the strict fsync policy, WAL write latency SLA is strictly guaranteed under 5.0 milliseconds. Group commit batching coalesces up to 32 concurrent writes per fsync cycle.

4. Bloom Filter Indexing

SSTable blocks include Block-level Bloom filters configured with 10 bits per key, achieving a target false positive rate of 1.0%. This prevents unnecessary disk seeks on point lookups for missing keys.